

<i>Work on part 3 and 4</i>						
Team Members	Role/ Responsibility	Email				
Innocent Tito Muvunyi		i.muvunyi@alustudent.com				
Linda Umwali		l.umwali@alustudent.com				
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Task Allocation and Tracker						
	Task Allocated	Assigned Member	Deadline	Completion Status (Not Started/In Progress/Completed/Did not participate)	Reviewed by Team? (Yes/No)	Comments
	Created em_algorithm.py file and the dataset	Patrick	02/07/2026	Completed	Yes	Initial file setup for Part 1
	Added necessary imports, prepared CSV data, created mixed heights list	Agnes	02/07/2026	Completed	Yes	Loaded GaltonFamilies dataset
	Added Gaussian probability function and initial guesses for the EM algorithm	Linda	02/07/2026	Completed	Yes	Set up mu1, mu2, var1, var2, pi1, pi2
	Added the E-step code, set max iterations, added tolerance check	Innocent	02/07/2026	Completed	Yes	Convergence logic for EM loop
	Added the M-step of the algorithm, random test height, output formatting	Patrick	02/07/2026	Completed	Yes	Classification demo output
	Added the .ipynb Jupyter file for the EM algorithm	Patrick	02/07/2026	Completed	Yes	Notebook version for presentation
	Created bayesian_prob file, added CSV prep code (IMDB dataset), initialized counters	Patrick	03/07/2026	Completed	Yes	Dataset for sentiment analysis
	Added chosen keywords, calculated prior, started Bayes' theorem loop	Innocent	03/07/2026	Completed	Yes	Scansdataset for keyword counts
	Added calculations for Marginal and Likelihood	Agnes	03/07/2026	Completed	Yes	P(keyword), P(keyword Positive)
	Added the posterior part and final result print formatting	Linda	03/07/2026	Completed	Yes	Bayes' Theorem output
	Manual gradient descent; initial prediction and first iteration	Patrick	04/07/2026	Completed	Yes	Handwritten calculations
	Manual gradient descent; second iteration	Linda	04/07/2026	Completed	Yes	Handwritten calculations
	Manual gradient descent; third iteration	Agnes	04/07/2026	Completed	Yes	Handwritten calculations
	Manual gradient descent; fourth iteration	Innocent	04/07/2026	Completed	Yes	Handwritten calculations

	Added grad descent section	Patrick	04/07/2026	Completed	Yes	Set up Part 4 file structure
	Added all imports, initial setup, feature/parameter definitions, plotting variables	Linda	04/07/2026	Completed	Yes	X, Y, m, b, alpha, history lists
	Committing cost function, derivative function, computed initial state	Agnes	04/07/2026	Completed	Yes	SciPy approx_fprime used
	Adding the update loop, final cost calculations, final predictions	Innocent	04/07/2026	Completed	Yes	Iterative GD updates with matrix ops
	Added plots for parameter changes over iterations and cost function	Patrick	05/07/2026	Completed	Yes	Matplotlib plots for m, b, error
	Added plot results, edited README, changed folder structure	All memebers	05/07/2026	Completed	Yes	inal cleanup and documentation

Meeting Attendance Log

Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps	
01/07/2026	Read and understand the assignment, as well as schedule other following meetings	[Nmae]	Patric, Agnes, Innocent, Linda	We discussed the assignment overview and understood what the assignment requirements	Create a GitHub repo, copy the spreadsheet contribution tracker, and make more research on part 1 of the assignment	
02/07/2026	Work on part 1 of the assignment	[Name]	Patric, Agnes, Innocent, Linda	Understood and started to implement Expectation-Maximization, we also finished drafting codes for part 1	Read, understand and work on part 2	
03/07/2026	Work on part 2	[Name]	Patric, Agnes, Innocent, Linda	Discussed keyword selection for sentiment analysis and Bayes' Theorem implementation; assigned prior/likelihood/posterior calculations	Finish part 2 code, start manual gradient descent calculations for part 3	
04/07/2026	Work on part 3 and 4	[Name]	Patric, Agnes, Innocent, Linda	Split manual gradient descent iterations among all 4 members; reviewed SciPy derivative implementation for part 4	Combine all handwritten calculations into one PDF, finalize part 4 plots.	
05/07/2026	Final review and presentation prep	[Name]	Patric, Agnes, Innocent, Linda	Reviewed all four parts together, ran the EM code live to confirm tracking table and classification output print correctly, assigned presentation sections	Do a final run-through before submitting.	